ResQnet — Disaster Response & Map Engine Functional Documentation

Role-based functionality, operational workflows, and how the integrated Map Engine supports disaster response

1. What is ResQnet?

ResQnet is an offline-first disaster response and field coordination platform designed to help citizens, emergency coordinators, responders, and administrators operate from a **unified operational picture during a disaster**.

It addresses fragmented information during disasters: citizens may not know which roads are safe; coordinators may not know where incidents and responders are located; responders may receive locations without reliable routes; roads can become blocked; and Safety Hubs may become full or unavailable.

Core concept: ResQnet connects people, incidents, responders, roads, hazards, and Safety Hubs into one coordinated disaster-response system.

Primary roles: **Citizen**, **Responder / NGO**, **Emergency Coordinator**, and **Admin**.

2. Why the Map Engine is Important

The map is not simply a visual component of ResQnet. It is the **operational layer** through which disaster information becomes actionable.

The existing **routing-lab/** Map Engine provides real road-following routing, GPS, route safety, hazards, blocked roads, blocked road sections, Safety Hubs, route progress, direction arrows, responder navigation, off-route detection, rerouting, and arrival detection.

This allows ResQnet to connect a reported incident with the physical route required to respond to it.

3. Overall Operational Flow

```
RESQNET
|
+---------------+----------------+
| | |
CITIZEN COORDINATOR RESPONDER
| | |
+---------------+----------------+
|
MAP ENGINE
|
+---------------+----------------+
| | |
GPS ROUTING MAP DATA
|
+---------------+----------------+
| | |
Hazards Road Blocks Safety Hubs
|
v
Operational Response
```

4. Citizen

Purpose

The Citizen interface helps a person understand the situation, report an emergency, find a safe route, reach an available Safety Hub, and receive relevant emergency information. The interface remains simple and map-first.

Citizen map functions

• **My Location:** explicitly activate GPS and display the current position.

• **Set Start:** choose the starting location using GPS, map selection, or a draggable A marker.

• **Set Destination:** select a Safety Hub or other destination as B.

• **Safe Route:** calculate an actual road-following route and evaluate hazards and closures.

Citizen evacuation

```
Citizen
|
Current GPS
|
Nearby Safety Hubs
|
+-- FULL / CLOSED / UNVERIFIED -> not eligible
|
+-- AVAILABLE -> Safe Route
|
Citizen
```

Citizen SOS

```
Citizen
|
Press SOS
|
Current GPS
|
SOS Incident
|
Coordinator
|
Assign Responder
|
Mission
|
Responder
```

The SOS location is tied to the incident location. Citizen GPS and responder GPS remain separate states.

5. Emergency Coordinator

The Coordinator operates the disaster control room. The operational map shows active SOS, incidents, responders, missions, hazards, dangerous zones, blocked roads, blocked road sections, and Safety Hubs.

Incident management

```
SOS
|
Coordinator Map
|
Incident appears
|
Review incident
|
Select responder
|
Create / assign mission
|
Responder dispatched
```

Hazard management

Coordinators can create or manage hazards such as flood, waterlogging, fire, debris, fallen tree, structural damage, and high-risk zones. Hazards become part of route safety evaluation.

```
Hazard created
|
Map Engine
|
Route evaluation
|
Intersects hazard?
|
```

```
YES
|
Unsafe / warning
|
Alternative route evaluated
```

Road blocking

A Coordinator can block an entire road. The Map Engine then avoids that road when calculating routes.

Road-section blocking

Only part of a road may be unusable. The Coordinator can select a road section using Closure Start and Closure End handles on the actual road geometry. The section between those gates becomes blocked.

Safety Hubs

Coordinators can create/manage Safety Hubs with location, optional entrance, capacity, occupancy, verification, active status, and operational capabilities such as shelter, food, water, medical support, sanitation, power/charging, registration/help desk, and special-needs support.

6. Responder / NGO

The Responder interface is an operational navigation interface focused on location, mission, route, changing road conditions, off-route detection, and arrival.

Live location

```
Live Location ON
|
Browser GPS
|
Responder live location
|
Map + Coordinator
```

Mission navigation

```
Coordinator
|
Mission
|
Responder
|
A = Responder GPS
B = Mission Target
|
OSRM + route safety + hazards + closures
|
Road-following route
|
Navigation
```

Direction arrows

```
FULL ROUTE
A -> -> -> -> -> -> -> B

GPS moves
-> -> -> -> B

Further progress
-> -> B

Arrival
no arrows
```

Direction arrows should represent the remaining route, not the route already travelled.

Off-route and rerouting

```
Navigation
|
GPS deviates
|
OFF ROUTE
|
Current GPS becomes new A
|
Mission target remains B
|
New route
|
Navigation continues
```

7. How the Map Engine Contributes to Rescue

1. Location Awareness

Shows where citizens, responders, incidents, Safety Hubs, hazards, and blocked roads are located.

2. Route Intelligence

Converts A → B into an actual road-following route.

3. Disaster-Aware Routing

Considers hazards, blocked roads, blocked road sections, route safety, and alternatives.

4. Dynamic Navigation

Provides GPS, route progress, direction arrows, off-route detection, rerouting, and arrival.

5. Coordination

Provides shared geographic context connecting Citizen, Coordinator, and Responder workflows.

8. Complete Rescue Workflow

```
CITIZEN
|
| SOS
v
INCIDENT
|
v
COORDINATOR
|
+---------+---------+
| |
Assess Incident Check Responders
| |
+---------+---------+
|
v
ASSIGN MISSION
|
v
RESPONDER
|
Live GPS ON
|
v
Navigate to Mission
|
v
MAP ENGINE
|
+--------------+--------------+
| | |
Roads Hazards Closures
```

```
| | |
+--------------+--------------+
|
v
Safe/Usable Route
|
v
Navigation
|
+--------+--------+
| |
ON ROUTE OFF ROUTE
| |
| REROUTE
+--------+--------+
|
v
ARRIVAL
|
v
ON SCENE
|
v
RESCUE OPERATION
```

9. Role-Based Functionality

Function	Citizen	Coordinator	Responder
View map	✓	✓	✓
GPS	✓	—	✓
My Location	✓	Optional	✓
A → B routing	✓	Optional	✓
Safe route	✓	✓	✓
Hazards visibility	✓	✓	✓
Create hazard	—	✓	—
View road closures	✓	✓	✓
Block entire road	—	✓	—
Block road section	—	✓	—
View Safety Hubs	✓	✓	✓
Create Safety Hub	—	✓	—
SOS	✓	Receives	—
View incidents	—	✓	Assigned
View responders	—	✓	Own location
Dispatch responder	—	✓	—
Receive mission	—	Creates	✓
Navigate to mission	—	—	✓
Route progress	✓	—	✓
Direction arrows	✓	—	✓
Off-route detection	✓	—	✓
Rerouting	✓	—	✓

Function	Citizen	Coordinator	Responder
Arrival detection	✓	—	✓

10. Final Product Concept

ResQnet is an offline-first disaster response and field coordination platform that creates a unified operational picture for citizens, coordinators, and responders, while its integrated Map Engine converts live geographic conditions into actionable, disaster-aware routing and navigation.

```
INFORMATION
|
Incident
|
DECISION
|
Coordinator dispatch
|
GEOGRAPHIC INTELLIGENCE
|
Map Engine
|
SAFE / USABLE ROUTE
|
NAVIGATION
|
RESPONDER
|
RESCUE
```

Core idea: the Map Engine is not just showing where a disaster is; it helps determine how people and responders can move through a changing disaster environment.

11. Map Engine Functional Foundation

The integrated Map Engine is based on the existing **routing-lab/** implementation. Its major capabilities include road-following routing, GPS/live location, route safety evaluation, hazards, full-road and road-section blocking, Safety Hubs and entrance routing, route progress, direction arrows, responder mission navigation, off-route detection, rerouting, and arrival detection.